(4)Evidence Preservation:

Evidence preservation ensures that digital evidence is collected, stored, and documented in a way that maintains its integrity and authenticity. This involves both volatile data collection and evidence collection using Velociraptor. Evidence hashed SHA-256 to verify integrity.

- Install velociraptor from official site and goto downloads and run command prompt

  velociraptor.exe gui

  then it opens browser

  http://127.0.0.1:8889

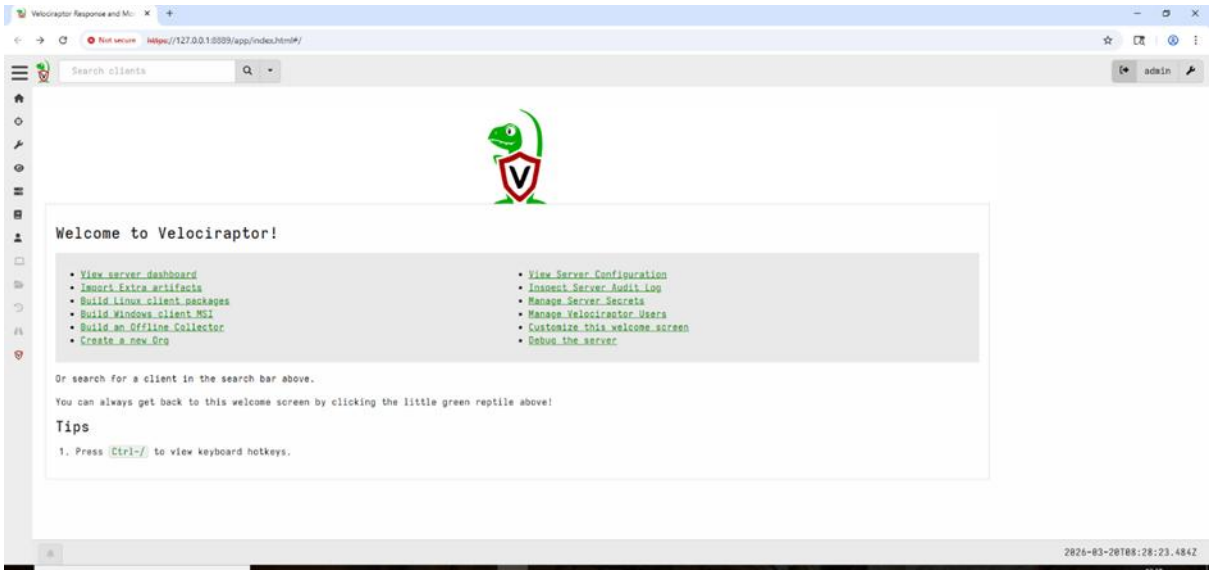


Figure 1: velociraptor browser

Volatile Data Collection:

In velociraptor

goto notebook → click new notebook → give name : netstat investigation → click create

then ,

add query cell → add cell → select VQL

edit cell → here paste

SELECT * FROM netstat()

Click save and run

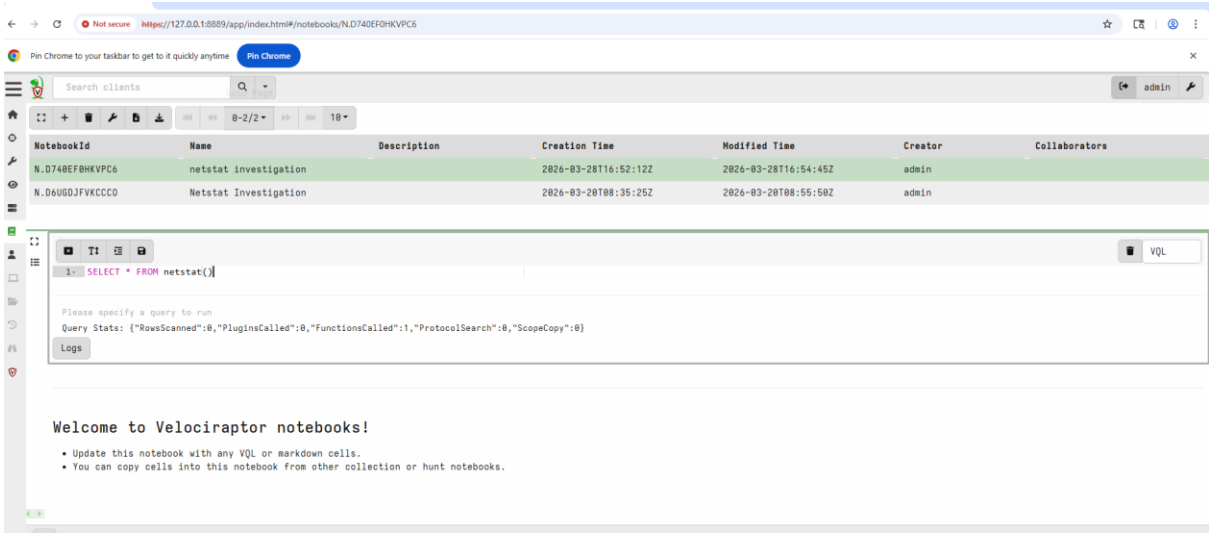


Figure 2: running SELECT * FROM netstat() to collect network connections from a Windows VM

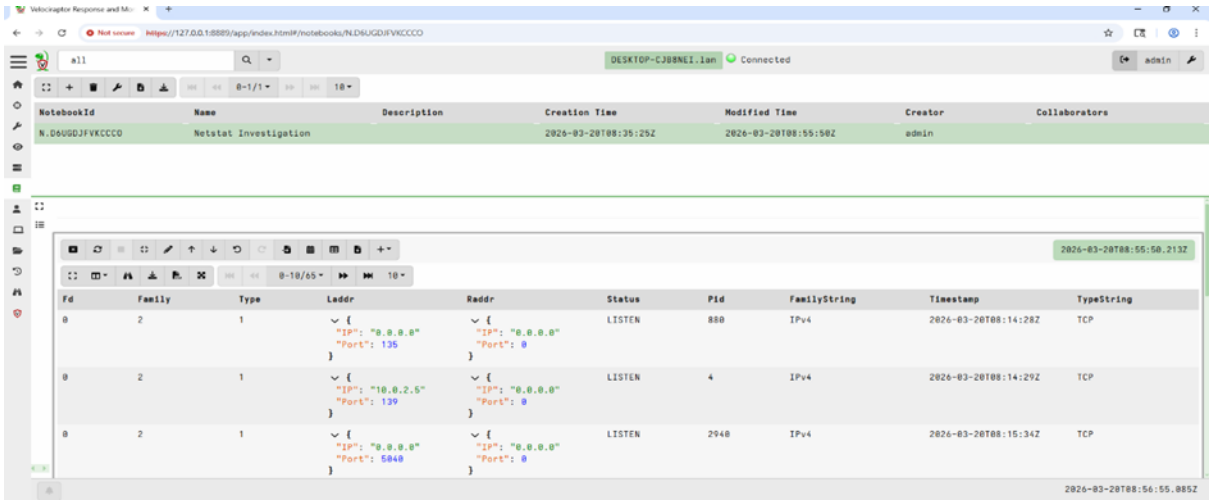


Figure 3

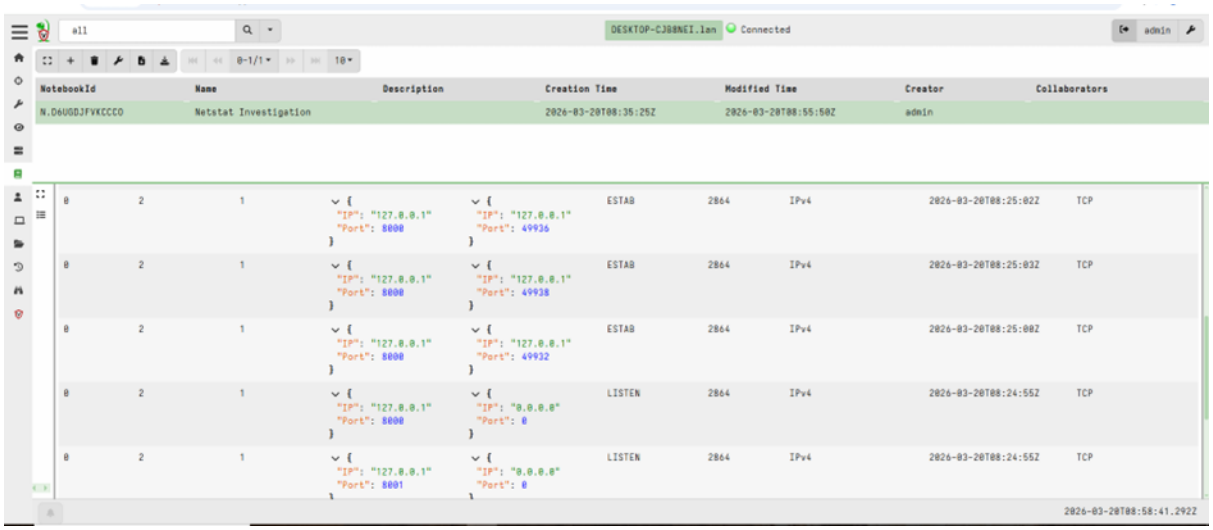


Figure 4

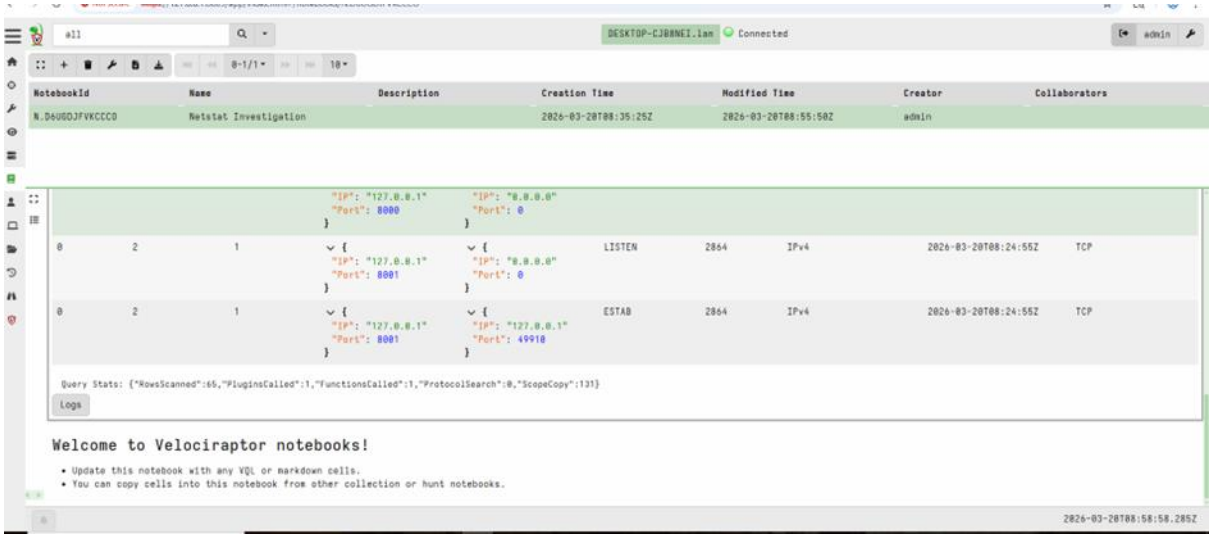
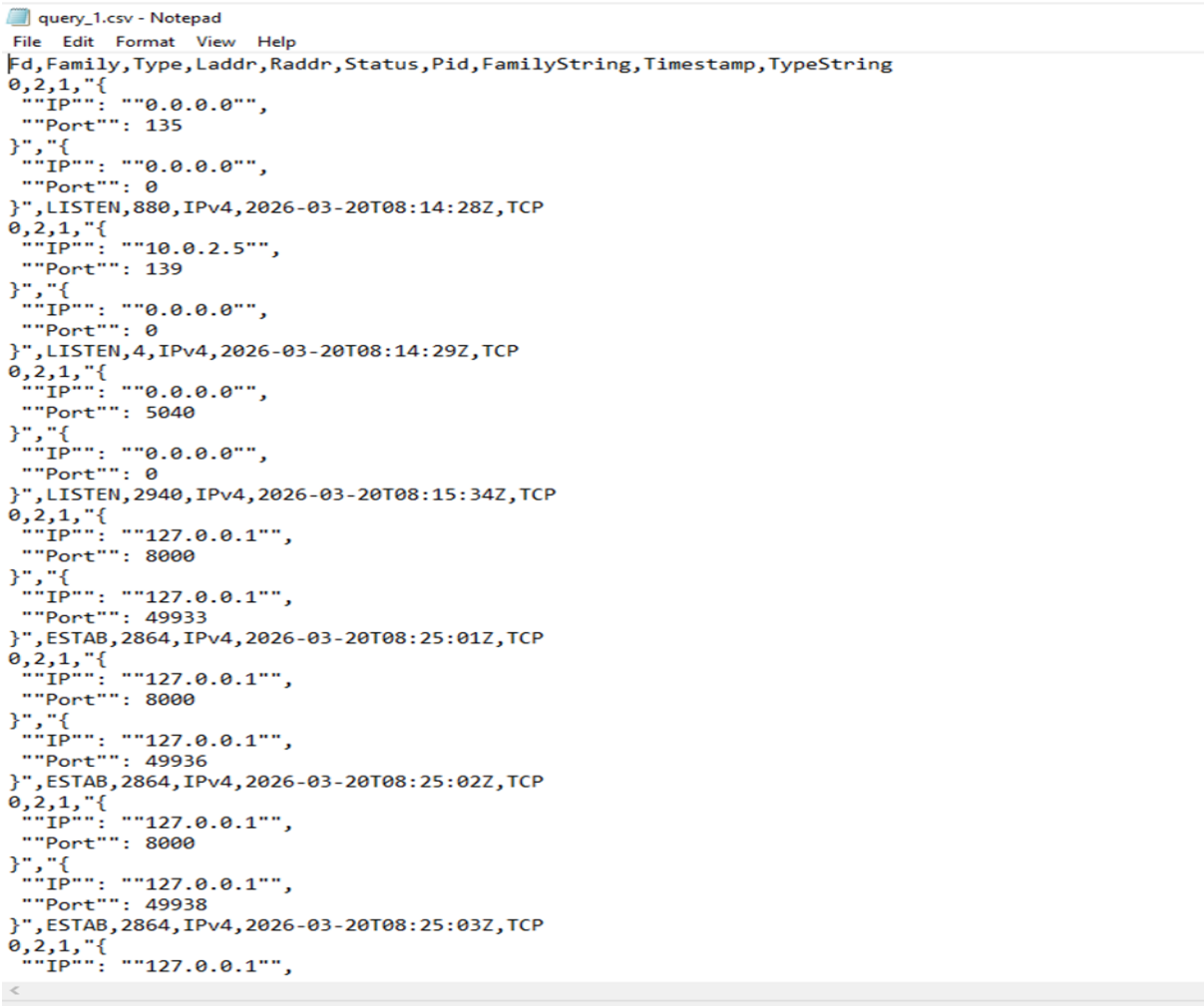


Figure 5

Figure 3,4,5 is a collection of Volatile Data

Export to csv → click on download csv



query_1.csv - Notepad

File Edit Format View Help

```
Fd,Family,Type,Laddr,Raddr,Status,Pid,FamilyString,Timestamp,TypeString
0,2,1,"{
 ""IP"": ""0.0.0.0"",
 ""Port"": 135
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,880,IPv4,2026-03-20T08:14:28Z,TCP
0,2,1,"{
 ""IP"": ""10.0.2.5"",
 ""Port"": 139
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,4,IPv4,2026-03-20T08:14:29Z,TCP
0,2,1,"{
 ""IP"": ""0.0.0.0"",
 ""Port"": 5040
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,2940,IPv4,2026-03-20T08:15:34Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49933
}",ESTAB,2864,IPv4,2026-03-20T08:25:01Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49936
}",ESTAB,2864,IPv4,2026-03-20T08:25:02Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49938
}",ESTAB,2864,IPv4,2026-03-20T08:25:03Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
```

Figure 6: CSV file of Volatile Data

Memory Dump Collection:

Goto host information → click on collect new articat → in serach bar type memory and select windows.memory.acquisition → click on launch

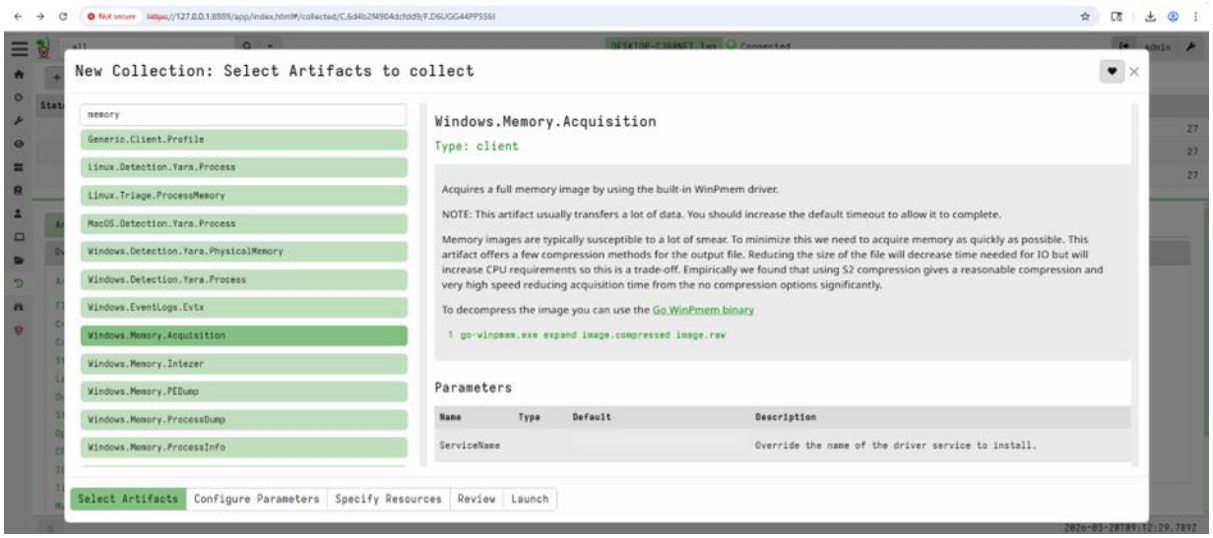


Figure 7: selecting artifacts to collect

After selecting launch, memory acquisition starts. When memory acquisition completed click on download and select prepare to csv then it downloads and also creates hash.

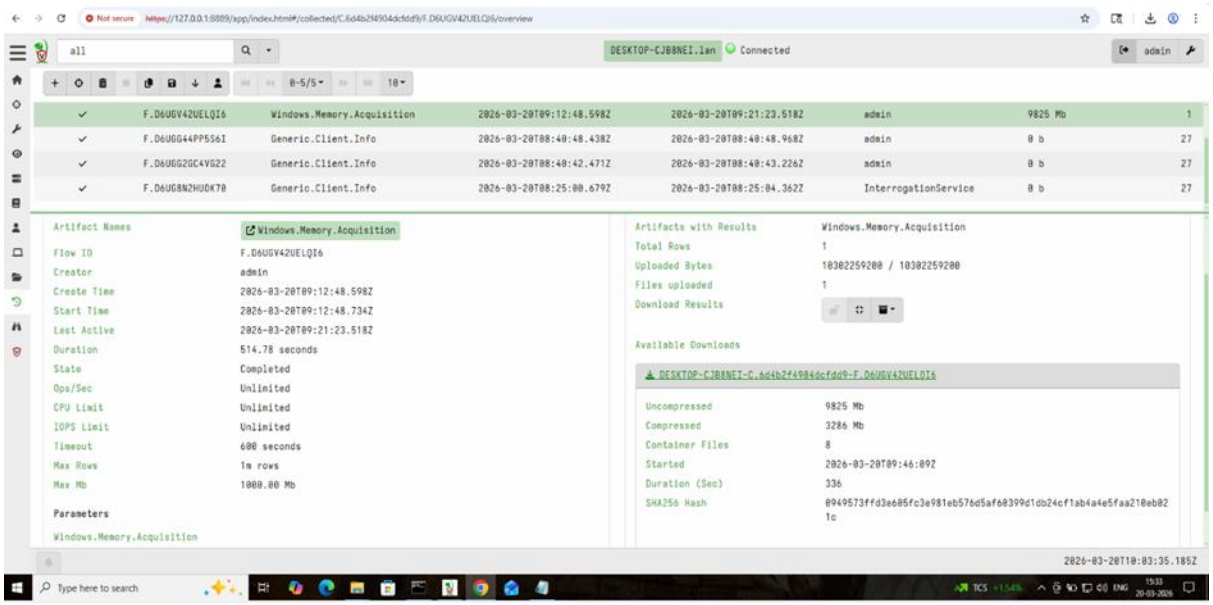


Figure 8: memory dump collection

Hash of memory dump is

0949573ffd3e605fc3e981eb576d5af60399d1db24cf1ab4a4e5faa210eb021c

Install memory dump as shown in below screenshot

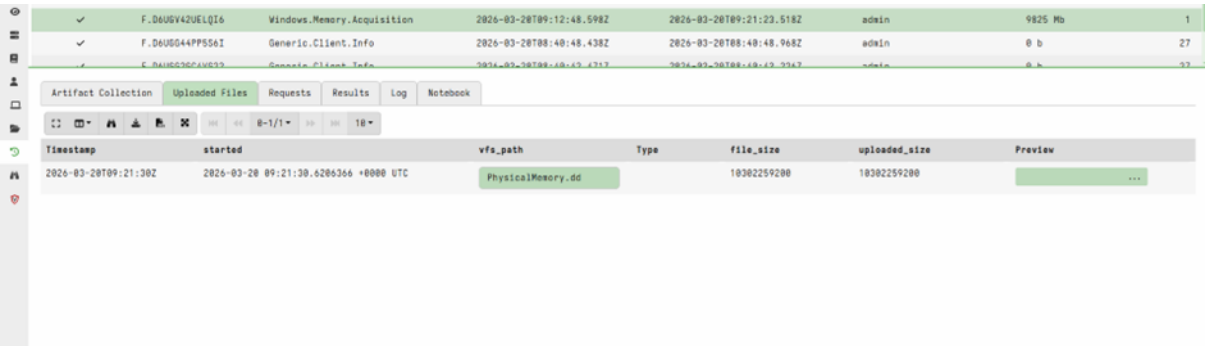


Figure 9: after downloading this , hash it

Open powershell and execute

Get-FileHash PhysicalMemory.dd -Algorithm SHA256

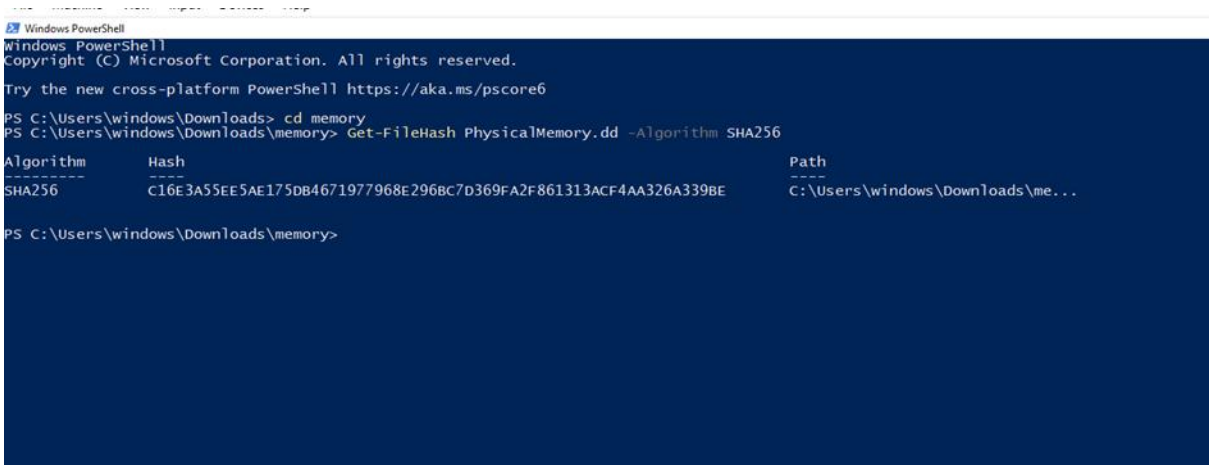


Figure 10: hash of the PhysicalMemory.dd

Document:

Item	description	Collected by	Date	Hash value
Memory dump	Server-X Dump	SOC analyst	20-02-2026	0949573ffd3e605fc3e981eb576d5af60399d1db24cf1ab4a4e5faa210eb021c
Volatile Data	Server-X Dump	SOC analyst	20-02-2026	-

Summarizing findings:
On 20-02-2026, the SOC analyst collected key digital evidence from Server-X. A memory dump was captured and hashed using SHA-256, ensuring data integrity. Additionally, volatile data such as active network connections was collected from the Windows system. All items were documented for chain-of-custody purposes, preserving their authenticity for further analysis.

In this task, digital evidence was collected from a Windows system while ensuring integrity and proper documentation. Volatile data, such as active network connections, was captured using Velociraptor and saved to CSV for analysis. A memory dump was acquired using Velociraptor and hashed with SHA-256 to verify its authenticity. All collected items were recorded in a chain-of-custody log, detailing the item, description, collector, date, and hash value. This process ensures evidence remains unaltered and reliable for forensic investigation, reporting, and potential legal proceedings, demonstrating best practices in digital evidence handling.